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Roll No

EX-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Electro Magnetic Field Theory

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any missing data.

1. a) Convert the vector $\vec{F} = 3\hat{a}_x + 2\hat{a}_y - 4\hat{a}_z$ into spherical co-ordinate system and evaluate the value of $\vec{F}$ at $\left(5, \frac{\pi}{3}, \frac{\pi}{6}\right)$. 7
- b) State and prove Gauss divergence theorem. Find the divergence of the vector $\vec{A} = 3x^2\hat{a}_x + 5x^2y\hat{a}_y + xyz^3\hat{a}_z$ 7
2. a) State and prove magnetic boundary conditions. 7
- b) If a potential $V = 3x^2yz + 2Ay^3z$ 7
 - i) Determine 'A' so that Laplace equation is satisfied.
 - ii) Using value of A, find electric field at (3, 2, -1).
3. a) Determine magnetic field intensity due to finite length current element and on the axis of a circular loop. 7
- b) Explain Faraday's law for electromagnetic induction in differential and integral form. 7

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4. a) State and explain Biot-Savart's law. Derive magnetic field intensity due to infinitely long wire carrying current 'I'. 7
b) Derive the mathematical expression for Poynting theorem. 7
5. a) Explain all forms of Maxwell's equation in time varying conditions and its physical significance. 7
b) Two point charges -4 nC and 5 nC are placed at $(2, -1, 3)$ and $(0, 4, -2)$ respectively. Find the potential at $(1, 0, 1)$ assuming zero potential at infinity. Right angle triangle, find the electric forces at the corner of the triangle. 7
6. a) Explain: 7
i) Displacement current
ii) Self and mutual inductance
b) Find the value of α and β for good conductors. Show that angle of characteristic impedance is always 45° for good conductors. 7
7. a) Derive an expression for attenuation constant, propagation constant and intrinsic impedance of an EM wave when its propagating through a lossy dielectric medium. 7
b) Write a short note on: 7
i) Continuity equation
ii) Relaxation time
iii) Resistance of conductor
8. Write the short note on (any three): 14
i) Reflection co-efficient and transmission coefficient
ii) Transmission line equation
iii) Propagation constant of transmission line
iv) Standing wave ratio
v) Distortion less and Loss less line.

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